

ASSIGNMENT

- 1 Both the end digits of a 99-digit number N are 2. N is divisible by 11, then all the middle digits are

Option:

- (a) 1 (b) 2 (c) 3 (d) 4

Answer: (d)

Solution:

A number is divisible by 11 if the difference of the sum of digits at odd and even places be either zero or multiple of 11.

If the middle digit be 4, then 24442 or 244442 etc. divisible by 11.

- 2 It is given that $(2^{32} + 1)$ is exactly divisible by a certain number. Which one of the following is also definitely divisible by the same number?

Option:

- (a) $2^{96} + 1$ (b) 7×2^{33} (c) $2^{16} - 1$ (d) $2^{16} + 1$

Answer: (a)

Solution:

$$\begin{aligned} 2^{96} + 1 &= (2^{32})^3 + 1^3 \\ &= (2^{32} + 1)(2^{64} - 2^{32} + 1) \end{aligned}$$

- 3 $4^{61} + 4^{62} + 4^{63} + 4^{64}$ is divisible by

Option:

- (a) 3 (b) 10 (c) 11 (d) 13

Answer: (b)

Solution:

$$\begin{aligned} 4^{61} + 4^{62} + 4^{63} + 4^{64} \\ &= 4^{61}(1 + 4 + 4^2 + 4^3) \\ &= 4^{61} \times 85, \text{ which is a multiple of 10} \end{aligned}$$

- 4 Which one of the numbers is divisible by 25?

Option:

- (a) 303310 (b) 373355 (c) 303375 (d) 22040

Answer: (c)

Solution:

A number is divisible by 25 if the last two digits are divisible by 25 or zero

- 5 A number when divided by 44, gives 432 as quotient and 0 as remainder. What will be the remainder

when dividing the same number by 31?

Option:

- (a) 3 (b) 4 (c) 5 (d) 6

Answer: (c)

Solution:

$$N = 44 \times 432 = 19008$$

$$\frac{N}{31} = \frac{19008}{31}$$

5 As remainder

- 6 The value of the digit d for which the number d456d is divisible by 18 is

Option:

- (a) 3 (b) 4 (c) 6 (d) 9

Answer: (c)

Solution:

d456d → divisible by 18

⇒ divisible by 2 and 9

To be divisible by 2, d must be even

d = 0, 2, 4, 6, 8

No. to be divisible by 9, sum of digits are divisible by 9.

⇒ d + 4 + 5 + 6 + d divisible by 9

⇒ 6 + 2d → divisible by 9

Put d = 4

6 + 8 → Not divisible by 9

Put d = 6

6 + 12 → divisible by 9

- 7 A number divided by 14 gives a remainder 8. What is the remainder, if the number is divided by 7?

Option

- (a) 1 (b) 2 (c) 3 (d) 4

Answer: (a)

Solution:

$$N = 14k + 8$$

$$\frac{N}{7} = \frac{14k + 8}{7} = \frac{14k + 7 + 1}{7}$$

⇒ Remainder = 1

- 8 Which of the following number is divisible by 99?

Option:

- (a) 3572404 (b) 135792 (c) 913464 (d) 114345

Answer: (d)

Solution:

For a number to be divisible by 99 it must be divisible by 9 and 11.

Now from given options,

Option (B), (C), (D) are divisible by 9

Now Let's check divisible by 11

(B) 135792

$$(2 + 7 + 2) - (9 + 5 + 1)$$

$$= 12 - 15 \text{ Not divisible by 11}$$

(C) 913464

$$(4 + 4 + 1) - (6 + 3 + 9)$$

Not divisible by 11

(D) 11345

$$(5 + 3 + 1) - (4 + 4 + 1) = 0$$

Divisible by 11

So D is answer

- 9 There is an N digit number ($N > 1$). If the sum of digits is subtracted from the number then the resulting number will be divisible by:

Option:

(a) 7

(b) 2

(c) 11

(d) 9

Answer: (d)

Solution:

Any number of N digits can be written as

$$a + 10b + 100c + 1000d + \dots$$

No. if we subtract sum of digits

$$\Rightarrow 0 + 9b + 99c + 999d + \dots$$

Clearly No. is divisible 9

Hence (D) is Answer

- 10 A number consists of two digits. If the number formed by interchanging the digit is added to the original number, the resulting number (i.e. the sum must be divisible by)

Option:

(a) 11

(b) 9

(c) 5

(d) 3

Answer: (a)

Solution:

Let the unit digit of the no. be x and ten's place be y

$$\text{Number} = 10y + x$$

Interchange digit's

$$= 10x + y$$

Adding result

$$= 10y + x + 10x + y = 11x + 11y = 11(x + y)$$

$\therefore$ Number is divisible by 11